

C1 protect the transformer from the noise from the mains. Although R1, R2 and C1 are not strictly necessary for the lamp, they should not be removed, especially in noisy urban or industrial environment.

The power for the LEDs is controlled with switch S2. In position 4 full power supply is applied, whereas in other positions (3, 2 and 1) the power and the light will be lower.

The LEDs are connected in a series and in couples in order to take the full power supply from the transformer as the forward voltage drop of LEDs is around half the secondary output voltage of the transformer. In one branch, the LEDs should be of the same type in order to have consistent forward voltage drop. The number of the branches of the LEDs depends on the type of the LEDs and transformer X1. The values of resistors R6 through R8 is selected based on the forward current rating of the LEDs.

After the selection of LEDs and transformer X1, simple calculations are needed to determine the values of current-limiting resistors and their power dissipations. The circuit does not require any adjustment and will start working immediately when switch S1 is switched on.

Construction and testing

An actual-size, single-side PCB for the AC-powered lamp is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on a PCB, enclose it in a suitable case. Fix all the LEDs in such a way that they stick out of the enclosed case. Use external wires to connect four-way rotary switch S2 to the PCB.

To test the circuit, power it on using switch S1 and keep switch S2 at position 4. Now measure voltages across the test points shown in the table.

Warning. The input section of this circuit is directly connected to

PARTS LIST

Semiconductors:

LED1-LED12 - 5mm white LED

Resistors (all 1/4-watt, 45% carbon):

R1 - 100-ohm, 0.5W

R2 - 100-ohm, 2W

R3 - 1-kilo-ohm, 1W

R4 - 2-kilo-ohm, 1W

R5 - 3.9-kilo-ohm, 2W

R6-R8 - 10-ohm

Capacitors:

C1 - 100nF, 630V polyester

C2 - 100nF ceramic disk

Miscellaneous:

X1 - 230V AC primary to 4.5V,

150mA secondary

transformer

S1 - On/off switch

S2 - 4-way rotary switch

CON1 - 2-pin connector

F1 - 150mA fuse

230V AC mains, which is potentially lethal! So take all necessary precautions while assembling the circuit! ●

The author worked as a researcher and assistant professor in Technical University of Sofia (Bulgaria) and as a lecturer and expert-lecturer in Kingdom of Morocco. Now he is an electronics engineer in the private sector in Bulgaria

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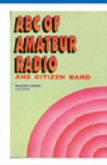
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